数学笔记

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Contents

Chapter 1

Symplectic Geometry

1.1 Hamiltonian Mechanics

Hamilton's Equations

Hamiltonian mechanics is a reformulation of classical mechanics. It arises from Lagrangian mechanics by a Legendre transformation. The state of a mechanical system is described by a point in a $2n$-dimensional phase space, with coordinates $(q_1, \ldots, q_n, p_1, \ldots, p_n)$, where the q_i are the generalized coordinates, and the p_i are the conjugate momenta. The time evolution of the system is given by **Hamilton's equations**:

$$\begin{cases} \dfrac{\partial q_i}{\partial t} = \dfrac{\partial H}{\partial p_i}, \\ \dfrac{\partial p_i}{\partial t} = -\dfrac{\partial H}{\partial q_i}, \end{cases} \quad i = 1, \ldots, n,$$

where $H(q, p, t)$ is the **Hamiltonian function**, which is usually the total energy of the system.

Suppoese $(q(t), p(t))$ is a solution to Hamilton's equations. Then for any function $f(q, p, t)$, its time evalution is given by

$$\frac{\mathrm{d}f}{\mathrm{d}t} = \frac{\partial f}{\partial t} + \sum_{i=1}^{n} \left(\frac{\partial f}{\partial q_i} \frac{\mathrm{d}q_i}{\mathrm{d}t} + \frac{\partial f}{\partial p_i} \frac{\mathrm{d}p_i}{\mathrm{d}t} \right) = \frac{\partial f}{\partial t} + \sum_{i=1}^{n} \left(\frac{\partial f}{\partial q_i} \frac{\partial H}{\partial p_i} - \frac{\partial f}{\partial p_i} \frac{\partial H}{\partial q_i} \right).$$

We define the **Poisson bracket** of two functions f and g by

$$\{f, g\} = \sum_{i=1}^{n} \left(\frac{\partial f}{\partial q_i} \frac{\partial g}{\partial p_i} - \frac{\partial f}{\partial p_i} \frac{\partial g}{\partial q_i} \right).$$

Then the time evolution of f can be written as

$$\frac{\mathrm{d}f}{\mathrm{d}t} = \frac{\partial f}{\partial t} + \{f, H\}.$$

So f is a constant of motion if and only if $\frac{\partial f}{\partial t} + \{f, H\} = 0$. If in particular f does not depend on t, then f is a constant of motion if and only if $\{f, H\} = 0$.

Hamilton's Equations and Symplectic Forms

To formulate Hamiltonian mechanics in a symplectic geometric framework, we introduce the symplectic form ω on the phase space:

$$\omega = \sum_{i=1}^{n} \mathrm{d}q_i \wedge \mathrm{d}p_i.$$

This is a closed, non-degenerate 2-form. Let $H = H(q, p)$ be a Hamiltonian function. The Hamiltonian vector field X_H associated to H is defined by the equation

$$\iota_{X_H}\omega = \mathrm{d}H,$$

where ι_{X_H} denotes the interior product with X_H. The left-hand side is

$$\iota_{X_H}\omega = \sum_{i=1}^{n} \Big(\mathrm{d}q_i(X_H)\mathrm{d}p_i - \mathrm{d}p_i(X_H)\mathrm{d}q_i\Big).$$

The right-hand side is

$$\mathrm{d}H = \sum_{i=1}^{n} \left(\frac{\partial H}{\partial q_i}\mathrm{d}q_i + \frac{\partial H}{\partial p_i}\mathrm{d}p_i\right).$$

So we have

$$\mathrm{d}q_i(X_H) = \frac{\partial H}{\partial p_i}, \quad \mathrm{d}p_i(X_H) = -\frac{\partial H}{\partial q_i}.$$

Thus the Hamiltonian vector field is

$$X_H = \sum_{i=1}^{n} \left(\frac{\partial H}{\partial p_i}\frac{\partial}{\partial q_i} - \frac{\partial H}{\partial q_i}\frac{\partial}{\partial p_i}\right).$$

The time evolution of the system is given by the flow of X_H, which is exactly Hamilton's equations.

For any smooth functions f, we can uniquely determine a verctor field X_f by solving the equation

$$\iota_{X_f}\omega = \mathrm{d}f.$$

For any two smooth functions f, g, We can define the Poission bracket to be

$$\{f, g\} := \omega(X_f, X_g).$$

When $\omega = \sum_{i=1}^n \mathrm{d}q_i \wedge \mathrm{d}p_i$, we can get

$$X_f = \sum_{i=1}^n \left(\frac{\partial f}{\partial p_i} \frac{\partial}{\partial q_i} - \frac{\partial f}{\partial q_i} \frac{\partial}{\partial p_i} \right).$$

So

$$\begin{aligned}
\omega(X_f, X_g) &= \sum_{i,j=1}^n \omega \left(\frac{\partial f}{\partial p_i} \frac{\partial}{\partial q_i} - \frac{\partial f}{\partial q_i} \frac{\partial}{\partial p_i}, \frac{\partial g}{\partial p_j} \frac{\partial}{\partial q_j} - \frac{\partial g}{\partial q_j} \frac{\partial}{\partial p_j} \right) \\
&= \sum_{i,j=1}^n \left(\frac{\partial f}{\partial q_i} \frac{\partial g}{\partial p_j} - \frac{\partial f}{\partial p_i} \frac{\partial g}{\partial q_j} \right) \delta_{ij} \\
&= \sum_{i=1}^n \left(\frac{\partial f}{\partial q_i} \frac{\partial g}{\partial p_i} - \frac{\partial f}{\partial p_i} \frac{\partial g}{\partial q_i} \right).
\end{aligned}$$

Canonical Transformations

Let $(Q(q,p), P(q,p))$ be a transformation from (q,p) to (Q,P). We say it is a **canonical transformation** if it preserves the form of Hamilton's equations, i.e., if $(Q(t), P(t))$ satisfies

$$\begin{cases} \dfrac{\partial Q_i}{\partial t} = \dfrac{\partial K}{\partial P_i}, \\ \dfrac{\partial P_i}{\partial t} = -\dfrac{\partial K}{\partial Q_i}, \end{cases} \quad i = 1, \dots, n,$$

for some Hamiltonian function $K(Q, P, t)$. This is equivalent to the **Poisson bracket condition**:

$$\begin{aligned}
\{P_i, Q_j\}_{q,p} &= \delta_{ij}, \\
\{Q_i, Q_j\}_{q,p} &= 0, \\
\{P_i, P_j\}_{q,p} &= 0.
\end{aligned}$$

Another equivalent condition is the **differential form condition**:

$$\sum_{i=1}^n \mathrm{d}Q_i \wedge \mathrm{d}P_i = \sum_{i=1}^n \mathrm{d}q_i \wedge \mathrm{d}p_i.$$

Canonical transformations and Symplectic Forms

Let $(Q(q,p), P(q,p))$ be a transformation from (q,p) to (Q,P). The Jacobian matrix of this transformation is

$$J = \frac{\partial(Q,P)}{\partial(q,p)} = \begin{pmatrix} \frac{\partial Q}{\partial q} & \frac{\partial Q}{\partial p} \\ \frac{\partial P}{\partial q} & \frac{\partial P}{\partial p} \end{pmatrix} =: \begin{pmatrix} A & B \\ C & D \end{pmatrix}.$$

Theorem 1.1.1. *The Poission bracket condition is equivalent to J^T being symplectic.*

Proof. The Poission bracket condition is

$$\begin{aligned}
\{P_k, Q_l\}_{q,p} &= \sum_{i=1}^{n} \Big(\frac{\partial Q_l}{\partial q_i}\frac{\partial P_k}{\partial p_i} - \frac{\partial Q_l}{\partial p_i}\frac{\partial P_k}{\partial q_i} \Big) = \delta_{kl}, \\
\{Q_k, Q_l\}_{q,p} &= \sum_{i=1}^{n} \Big(\frac{\partial Q_k}{\partial q_i}\frac{\partial Q_l}{\partial p_i} - \frac{\partial Q_k}{\partial p_i}\frac{\partial Q_l}{\partial q_i} \Big) = 0, \\
\{P_k, P_l\}_{q,p} &= \sum_{i=1}^{n} \Big(\frac{\partial P_k}{\partial q_i}\frac{\partial P_l}{\partial p_i} - \frac{\partial P_k}{\partial p_i}\frac{\partial P_l}{\partial q_i} \Big) = 0.
\end{aligned}$$

Written in a tigher form, they are

$$\begin{aligned}
\frac{\partial Q}{\partial q}\Big(\frac{\partial P}{\partial p}\Big)^T - \frac{\partial Q}{\partial p}\Big(\frac{\partial P}{\partial q}\Big)^T &= I \Rightarrow AD^T - BC^T = I, \\
\frac{\partial Q}{\partial q}\Big(\frac{\partial Q}{\partial p}\Big)^T - \frac{\partial Q}{\partial p}\Big(\frac{\partial Q}{\partial q}\Big)^T &= 0 \Rightarrow AB^T - BA^T = 0, \\
\frac{\partial P}{\partial q}\Big(\frac{\partial P}{\partial p}\Big)^T - \frac{\partial P}{\partial p}\Big(\frac{\partial P}{\partial q}\Big)^T &= 0 \Rightarrow CD^T - DC^T = 0.
\end{aligned}$$

This is equivalent to say

$$\begin{aligned}
J\Omega J^T &= \begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix} \begin{pmatrix} A^T & C^T \\ B^T & D^T \end{pmatrix} \\
&= \begin{pmatrix} AB^T - BA^T & AD^T - BC^T \\ CB^T - DA^T & CD^T - DC^T \end{pmatrix} \\
&= \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix} = \Omega.
\end{aligned}$$

□

Theorem 1.1.2. *The differential form condition is equivalent to J being symplectic.*

Proof. Let $\omega_{(q,p)} = \sum_{i=1}^{n} \mathrm{d}q_i \wedge \mathrm{d}p_i$, $\omega_{(Q,P)} = \sum_{i=1}^{n} \mathrm{d}Q_i \wedge \mathrm{d}P_i$. Then

$$\begin{aligned}
\omega_{(Q,P)} &= \sum_{i=1}^{n} \left(\sum_{k=1}^{n} \frac{\partial Q_i}{\partial q_k}\mathrm{d}q_k + \frac{\partial Q_i}{\partial p_k}\mathrm{d}p_k \right) \wedge \left(\sum_{l=1}^{n} \frac{\partial P_i}{\partial q_l}\mathrm{d}q_l + \frac{\partial P_i}{\partial p_l}\mathrm{d}p_l \right) \\
&= \sum_{i=1}^{n} \sum_{k,l=1}^{n} \left(\frac{\partial Q_i}{\partial q_k}\frac{\partial P_i}{\partial p_l} - \frac{\partial Q_i}{\partial p_l}\frac{\partial P_i}{\partial q_k} \right) \mathrm{d}q_k \wedge \mathrm{d}p_l \\
&\quad + \sum_{i=1}^{n} \sum_{k<l} \left(\frac{\partial Q_i}{\partial q_k}\frac{\partial P_i}{\partial q_l} - \frac{\partial Q_i}{\partial q_l}\frac{\partial P_i}{\partial q_k} \right) \mathrm{d}q_k \wedge \mathrm{d}q_l \\
&\quad + \sum_{i=1}^{n} \sum_{k<l} \left(\frac{\partial Q_i}{\partial p_k}\frac{\partial P_i}{\partial p_l} - \frac{\partial Q_i}{\partial p_l}\frac{\partial P_i}{\partial p_k} \right) \mathrm{d}p_k \wedge \mathrm{d}p_l.
\end{aligned}$$

Thus $\omega_{(Q,P)} = \omega_{(q,p)}$ if and only if

$$\begin{aligned}
\sum_{i=1}^{n} \left(\frac{\partial Q_i}{\partial q_k}\frac{\partial P_i}{\partial p_l} - \frac{\partial Q_i}{\partial p_l}\frac{\partial P_i}{\partial q_k} \right) &= \delta_{kl}, \\
\sum_{i=1}^{n} \left(\frac{\partial Q_i}{\partial q_k}\frac{\partial P_i}{\partial q_l} - \frac{\partial Q_i}{\partial q_l}\frac{\partial P_i}{\partial q_k} \right) &= 0, \\
\sum_{i=1}^{n} \left(\frac{\partial Q_i}{\partial p_k}\frac{\partial P_i}{\partial p_l} - \frac{\partial Q_i}{\partial p_l}\frac{\partial P_i}{\partial p_k} \right) &= 0.
\end{aligned}$$

Written in a tigher form, they are

$$\begin{aligned}
\left(\frac{\partial Q}{\partial q}\right)^T \frac{\partial P}{\partial p} - \left(\frac{\partial P}{\partial q}\right)^T \frac{\partial Q}{\partial p} &= I \Rightarrow A^T D - C^T B = I, \\
\left(\frac{\partial Q}{\partial q}\right)^T \frac{\partial Q}{\partial p} - \left(\frac{\partial P}{\partial q}\right)^T \frac{\partial P}{\partial p} &= 0 \Rightarrow A^T B - C^T A = 0, \\
\left(\frac{\partial Q}{\partial p}\right)^T \frac{\partial P}{\partial q} - \left(\frac{\partial P}{\partial p}\right)^T \frac{\partial Q}{\partial q} &= 0 \Rightarrow B^T D - D^T B = 0.
\end{aligned}$$

This is equivalent to say

$$\begin{aligned}
J^T \Omega J &= \begin{pmatrix} A^T & C^T \\ B^T & D^T \end{pmatrix} \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix} \begin{pmatrix} A & B \\ C & D \end{pmatrix} \\
&= \begin{pmatrix} A^T C - C^T A & A^T D - C^T B \\ B^T C - D^T A & B^T D - D^T B \end{pmatrix} \\
&= \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix} = \Omega.
\end{aligned}$$

□

Theorem 1.1.3. *J is symplectic if and only if J^T is symplectic.*

Proof.

$$\begin{aligned} J \text{ is symplectic} &\Leftrightarrow J^T \Omega J = \Omega \\ &\Leftrightarrow J^{-1}\Omega(J^T)^{-1} = \Omega \\ &\Leftrightarrow J\Omega J^T = \Omega \\ &\Leftrightarrow J^T \text{ is symplectic.} \end{aligned}$$

□

Those three theorems together give the following conclusion:

Theorem 1.1.4. *The Poission bracket condition is equivalent to the differential form condition.*